

LATENT CHANGE IN RECURRENT CHOICE DATA

ULF BÖCKENHOLT

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

ROLF LANGEHEINE

INSTITUTE FOR SCIENCE EDUCATION
UNIVERSITY OF KIEL

This paper introduces dynamic latent-class models for the analysis and interpretation of stability and change in recurrent choice data. These latent-class models provide a nonparametric representation of individual taste differences. Changes in preferences are modeled by allowing for individual-level transitions from one latent class to another over time. The most general model facilitates a saturated representation of class membership changes. Several special cases are presented to obtain a parsimonious description of latent change mechanisms. An easy to implement EM algorithm is derived for parameter estimation. The approach is illustrated by a detailed analysis of a purchase incidence data set.

Key words: latent class models, Poisson distribution, count data, EM algorithm, latent change, choice behavior.

Introduction

Modeling choice behavior over time is an intriguing methodological problem. We have to take into account not only that individuals differ in their preferences but also that they differ in the way they change their preferences. For example, current preferences may depend on such diverse factors as previous choices, interchoice times, or the length of time the current preference state has been occupied. Some consumers are brand-loyal and buy the same brand repeatedly over time, others are variety-seekers and switch among brands in some systematic or unsystematic fashion. Similarly, decision-makers may differ in their familiarity with the features of the choice options with the result that some decision-makers are stable in their preferences while others may change their preference according to some learning process. These examples illustrate that when modeling preferences over time, at the very least, we need to distinguish two sources of systematic variation: Heterogeneity caused by individual taste differences and structural dependencies between current and past preference states.

The approach proposed here is developed for (but not limited to) the analysis of recurrent choices. Such data are readily available in the form of repeated multi-brand purchases of consumer panels. In a recurrent choice situation, decision makers choose repeatedly among alternatives within some time period. The choice process at the individual level is described by the choice probabilities for the alternatives, and the rate by which choices are made. Because decision makers differ in their choice rates and probabilities, it is necessary to take variations of these parameters into account for accurate choice predictions on the aggregate level.

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Requests for reprints should be sent to Ulf Böckenholt, Department of Psychology, University of Illinois at Urbana-Champaign, Champaign, IL 61820.

Latent-class models have proven useful for representing preference heterogeneity because they facilitate a nonparametric approximation of the underlying distribution of the decision-makers' choice parameters (Böckenholt, 1993). According to these models, decision-makers are grouped with high intragroup similarity and large intergroup differences, and each group or class is characterized by its rate of making choices and its set of choice probabilities for the options. When the decision-makers' preferences are stable, the same procedure can be applied to choice data observed over several time periods. In most choice studies, however, a more flexible approach is needed that goes beyond the restrictive assumption of stable preferences and static class-membership over time.

The approach presented in this paper provides a straightforward solution to this problem. Instead of constraining class-membership to be static, the proposed set of models allows for systematic and unsystematic shifts in the latent-class positions and latent-class specific choice rates and probabilities. Thus, the models can accommodate decision-makers who switch from one preference state to another either according to some well-defined (e.g., learning) process or in an arbitrary fashion. As a result, we obtain a parsimonious account of preference change over time while taking into account (arbitrary) preference heterogeneity in the population of decision-makers. Our model framework builds on and generalizes the multinomial latent-class models for longitudinal data originally proposed by Lazarsfeld (1950) and Wiggins (1955, 1973).

The remainder of this paper is organized as follows: Section 1 presents the latent-class recurrent choice model for one time period. In section 2, we introduce a dynamic version of this latent-class model that allows for latent preference changes over time. Section 3 addresses estimation and model testing issues. The paper concludes with an application to a consumer panel data set.

The Recurrent Choice Model

We assume that data are collected in a recurrent choice study where decision makers choose repeatedly among choice alternatives over several specified time periods ($t = 1, \dots, T$). The number of selections of choice option i ($i = 1, \dots, I$) by decision-maker n ($n = 1, \dots, N$) within time period t is denoted by $x_{ni}^{(t)}$ and the corresponding total number of choices is $\sum_{i=1}^I x_{ni}^{(t)} = x_n^{(t)}$. An individual selection pattern is represented by $\mathbf{x}_n^{(t)} = (x_{n1}^{(t)}, \dots, x_{nI}^{(t)})$. Two components of recurrent choice data are distinguished for decision maker n , the total number of choices made within a specified time period, $x_n^{(t)}$, and the probability by which an alternative is chosen, $p_{ni}^{(t)}$. Both statistics, $x_n^{(t)}$ and $p_{ni}^{(t)}$ describe different aspects of the choice process, and, consequently may reflect the effects of different factors (Ehrenberg, 1988). In particular, it seems plausible to that changes in the choice options' attributes do not influence the rate by which decisions are made.

Choices are described by a multinomial process with probability vector $\mathbf{p}^{(t)} = (p_1^{(t)}, \dots, p_I^{(t)})$ and $\sum_{i=1}^I p_i^{(t)} = 1$,

$$\Pr_{\mathcal{M}}(\mathbf{x}_n^{(t)}; x_n^{(t)}, \mathbf{p}^{(t)}) = x_n^{(t)!} \prod_{i=1}^I \frac{p_i^{(t)^{x_{ni}^{(t)}}}}{x_{ni}^{(t)}!}.$$

The total number of choices, $x_n^{(t)}$, within time period t is described by a Poisson process with mean rate $\zeta^{(t)}$,

$$\Pr_{\mathcal{P}}(x_n^{(t)}; \zeta^{(t)}) = \frac{e^{-\zeta^{(t)}} \zeta^{(t)^{x_n^{(t)}}}}{x_n^{(t)}!}, \quad x_n^{(t)} = 0, 1, \dots$$

This assumption implies that the probability of making a choice is not affected by the length of the time since the last choice incidence. Although it is important to test the validity of this assumption in any particular application, there is considerable evidence, (for example, in the context of purchasing behavior) that the variability in individual choice frequencies is well described by a Poisson distribution (Kahn & Morrison, 1989).

Dzhafarov and Böckenholt (1995) showed that the Poisson distribution is the only distribution consistent with the assumption that the individual choice frequencies x_{ni} are independent when the conditional distribution of $\mathbf{x}_n^{(t)}$ given $x_n^{(t)}$ is multinomial. It follows from this independence assumption that the distribution of the selection pattern $\mathbf{x}_n^{(t)}$ is given by the product of the Poisson and the multinomial distribution,

$$\Pr_{\mathcal{MP}}(\mathbf{x}_n^{(t)}; \zeta^{(t)}, \mathbf{p}^{(t)}) = \Pr_{\mathcal{P}}(x_n^{(t)}; \zeta^{(t)}) \Pr_{\mathcal{M}}(\mathbf{x}_n^{(t)}; x_n^{(t)}, \mathbf{p}^{(t)}),$$

or, equivalently,

$$\Pr_{\mathcal{MP}}(\mathbf{x}_n^{(t)}; \zeta^{(t)}, \mathbf{p}^{(t)}) = \prod_{i=1}^I \Pr_{\mathcal{P}}(x_{ni}^{(t)}; \zeta^{(t)} p_i^{(t)}).$$

Possibly arbitrary assumptions about the underlying joint distribution of the choice rates and probabilities in the population can be avoided by adopting a latent-class approach (Böckenholt, 1993; Poulsen, 1982). In this case, the heterogeneous population of decision makers at time period t is represented by S latent classes with relative class size $\pi_s^{(t)}$ ($s = 1, \dots, S$). The latent classes are defined to be mutually exclusive and exhaustive, and

$$\sum_{s=1}^S \pi_s^{(t)} = 1, \quad t = 1, \dots, T.$$

Within each latent class the total number of choices, $x_n^{(t)}$, is described by a Poisson process and the distribution of the choices among the alternatives is defined by the multinomial model. The probability of observing a selection pattern $\mathbf{x}_n^{(t)}$ given that person n is a member of latent class s is,

$$\Pr_{\mathcal{MP}}(\mathbf{x}_n^{(t)}; \mathbf{p}_s^{(t)}, \zeta_s^{(t)}) = \Pr_{\mathcal{P}}(x_n^{(t)}; \zeta_s^{(t)}) \Pr_{\mathcal{M}}(\mathbf{x}_n^{(t)}; x_n^{(t)}, \mathbf{p}_s^{(t)}),$$

and $\mathbf{p}_s^{(t)} = (p_{1s}^{(t)}, \dots, p_{Is}^{(t)})$. Thus, each latent class is characterized by its choice probability vector, $\mathbf{p}_s^{(t)}$, and rate parameter, $\zeta_s^{(t)}$, at time period t . The marginal probability of observing a particular selection pattern is

$$\Pr_{\mathcal{MP}}(\mathbf{x}_n^{(t)}) = \sum_{s=1}^S \pi_s^{(t)} \Pr_{\mathcal{MP}}(\mathbf{x}_n^{(t)}; \mathbf{p}_s^{(t)}, \zeta_s^{(t)}).$$

The next section discusses various generalizations of this model that account for time-dependent changes in class membership and in the class-specific choice rates and probabilities.

¹ To simplify notation, we do not list the class-specific parameters, $\mathbf{p}_s^{(t)}$, $\zeta_s^{(t)}$, $\pi_s^{(t)}$, ($s = 1, \dots, S$; $t = 1, \dots, T$) in this and the following expressions for the marginal probabilities.

Dynamic Latent Class Models

Each latent class represents a particular preference state characterized by a class-specific set of choice probabilities and rates. The restriction that latent-class membership is stable implies that decision-makers do not change their preferences over time. While this may be the case for some individuals, it is clear that, in general, this constraint is not appropriate. For example, variety-seeking, learning experiences with the choice options, or changes in person-specific attributes (e.g., demographics) may lead to systematic switches among classes.

The idea to model changes in class membership over time was first proposed by Wiggins (1955, 1973; see also T. W. Anderson, 1954). In the context of the multinomial latent-class models, Wiggins suggested that time changes in attitudes can be captured by a latent Markov process. Because of estimation difficulties, his work proved to be of little consequence until Poulsen (1982, 1990) showed that the latent Markov model can be estimated by the EM algorithm (Dempster, Laird, & Rubin, 1977). Further detail and extensions of the multinomial latent Markov model can be found in Langeheine and van de Pol (1990). Of course, the latent Markov model may not always be appropriate to describe changes in class-membership over time. Beginning with the fundamental work of Lazarsfeld (1950), the list of possible latent change mechanisms was extended by the adoption of loglinear constraints suitable for the investigation of time-dependent changes in class-membership (E. B. Andersen, 1985; Clogg, in press; Goodman, 1974; Haberman, 1977; Hagenaars, 1990). In particular, Clogg (in press) and Hagenaars (1990) provide valuable overviews about recent developments in this area.

The application of this work to the multinomial-Poisson model is straightforward. However, because the multinomial-Poisson latent-class model has better identifiability properties² than the multinomial latent-class model (Johnson, Kotz, & Kemp, 1992), it facilitates the implementation and testing of a wider range of latent-change mechanisms. In particular, even when there is only one option during each time period it is possible to estimate the saturated (i.e., unconstrained) latent-change model and to compare it to more restrictive versions in a hierarchical fashion.

Our classification of time-dependent latent-class models contrasts the no-change case with the cases of systematic and unsystematic changes in class-memberships, choice probabilities and rates over time for a given number of classes. The most general representation assumes changes in latent-class memberships, choice rates and probabilities from one time period to the next. To simplify notation, we present the marginal probability for three time periods and restrict the number of latent classes to be equal over time. In this case, the dynamic, time-inhomogeneous latent class model is given by

$$\Pr(\mathbf{x}_n^{(1)}, \mathbf{x}_n^{(2)}, \mathbf{x}_n^{(3)}) = \sum_{a,b,c} \pi_{abc} \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(1)}; \mathbf{p}_a^{(1)}, \zeta_a^{(1)}) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(2)}; \mathbf{p}_b^{(2)}, \zeta_b^{(2)}) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(3)}; \mathbf{p}_c^{(3)}, \zeta_c^{(3)}), \quad (1)$$

where π_{abc} refers to the probability of belonging to classes, a , b , and c at time points (1), (2), and (3), respectively. We allow for time-inhomogeneous choice rates and probabilities because in some applications known attributes of the choice options and the decision-makers may change over time. To simplify the interpretation of the class-specific parameters in this case, it may prove useful to functionally relate them to these

² The identifiability of the multinomial-Poisson model follows from the fact that it can be equivalently represented as a product-Poisson mixture. Teicher (1961, 1967) showed that product-Poisson mixtures are identifiable.

attributes. In the empirical application section of this paper, we illustrate this approach by expressing the class-specific choice probabilities as a function of the brands' prices which vary over time.

Depending on whether we expect time-dependent changes in the class size parameters or the choice rates and probabilities, (1) can be simplified. If both the class sizes and class-specific probabilities and rates are equal over time, we obtain the static latent-class multinomial-Poisson model

$$\Pr(\mathbf{x}_n^{(1)}, \mathbf{x}_n^{(2)}, \mathbf{x}_n^{(3)}) = \sum_s \pi_s \prod_{t=1}^3 \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(t)}; \mathbf{p}_s, \zeta_s). \quad (2)$$

Thus, each latent class is characterized by time-homogeneous choice rates and probabilities. Although this representation can account for heterogeneity in preferences, it cannot accommodate preference changes for a given number of latent classes. This limitation is overcome by the following expressions. If the class-specific probabilities and/or rates vary over time, we obtain

$$\Pr(\mathbf{x}_n^{(1)}, \mathbf{x}_n^{(2)}, \mathbf{x}_n^{(3)}) = \sum_s \pi_s \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(1)}; \mathbf{p}_s^{(1)}, \zeta_s^{(1)}) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(2)}; \mathbf{p}_s^{(2)}, \zeta_s^{(2)}) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(3)}; \mathbf{p}_s^{(3)}, \zeta_s^{(3)}), \quad (3)$$

and, if members switch between latent classes, we get

$$\Pr(\mathbf{x}_n^{(1)}, \mathbf{x}_n^{(2)}, \mathbf{x}_n^{(3)}) = \sum_{a,b,c} \pi_{abc} \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(1)}; \mathbf{p}_a, \zeta_a) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(2)}; \mathbf{p}_b, \zeta_b) \Pr_{\mathcal{M}\mathcal{P}}(\mathbf{x}_n^{(3)}; \mathbf{p}_c, \zeta_c). \quad (4)$$

Provided the latent change mechanism is stationary, we can use Model (4) to predict choice behavior at future time periods.

Many useful special cases of Models (1) and (4) can be derived by imposing constraints on π_{abc} to describe switches among latent classes. For example, an important representation for latent change is the first-order Markov chain with stationary transition probabilities. This restriction was originally proposed by Wiggins (1955, 1973)

$$\pi_{abc} = \pi_a^{(1)} \pi_{b|a} \pi_{c|b}, \quad (5)$$

where $\pi_{b|a}$ refers to the probability of being a member of class b given membership in class a during the previous time period. The static latent-class model is obtained by setting $\pi_{b|a} = 0$ and $\pi_{c|b} = 0$ when $a \neq b$ and $b \neq c$, respectively. Other constraints can be implemented by imposing a linear or loglinear structure on π_{abc} (Bishop, Fienberg, & Holland, 1975; Hagenaars, 1990). For instance, a special case of (5) is the random change model

$$\pi_{abc} = \pi_a \pi_b \pi_c,$$

which describes time-dependent changes as a function of the class sizes which, in turn, are constrained to be equal over time.

A third restriction that will prove important in the application section of this paper is latent symmetry. Under this constraint, an equal number of decision-makers

Table 1.

Representation of a first-order, stationary two-class Markov model
as a no-change, eight-class model

Class	Parameter constraints	
	Class size	Rate parameters
1	$\pi_a^{(1)} \pi_a a \pi_a a$	$\zeta_a^{(1)}, \zeta_a^{(2)}, \zeta_a^{(3)}$
2	$\pi_a^{(1)} \pi_a a \pi_b a$	$\zeta_a^{(1)}, \zeta_a^{(2)}, \zeta_b^{(3)}$
3	$\pi_a^{(1)} \pi_b a \pi_a b$	$\zeta_a^{(1)}, \zeta_b^{(2)}, \zeta_a^{(3)}$
4	$\pi_a^{(1)} \pi_b a \pi_b b$	$\zeta_a^{(1)}, \zeta_b^{(2)}, \zeta_b^{(3)}$
5	$\pi_b^{(1)} \pi_a b \pi_a a$	$\zeta_b^{(1)}, \zeta_a^{(2)}, \zeta_a^{(3)}$
6	$\pi_b^{(1)} \pi_a b \pi_b a$	$\zeta_b^{(1)}, \zeta_a^{(2)}, \zeta_b^{(3)}$
7	$\pi_b^{(1)} \pi_b b \pi_a b$	$\zeta_b^{(1)}, \zeta_b^{(2)}, \zeta_a^{(3)}$
8	$\pi_b^{(1)} \pi_b b \pi_b b$	$\zeta_b^{(1)}, \zeta_b^{(2)}, \zeta_b^{(3)}$

Note: *a* and *b* refer to the first and second class of the two-class Markov model, respectively. Superscripts refer to the three time periods.

switches from class *a* to class *b* as from class *b* to *a*. As a result, the order of the time periods is irrelevant for predicting changes in class membership.

A close inspection of Models (1) through (4) reveals that each dynamic latent-class model can be represented equivalently as a constrained static latent class model (Goodman, 1974; Lazarsfeld, 1950). For instance, in the case of three time periods a two-class model with latent change following a first-order, stationary Markov-chain is equivalent to a (no-change) eight-class model with certain parameter constraints. These constraints are given in Table 1 for the class sizes and the choice rate parameters for the three time periods. (The constraints for the choice probabilities follow the same pattern given for the choice rates.) This example demonstrates that without some knowledge about the true number of latent classes it is difficult to distinguish empirically between hypotheses of latent change and heterogeneity in preferences. Fortunately, in the case of the multinomial-Poisson model, we can obtain a bound on the number of latent classes by determining the number of classes at each of the *T* time periods separately. For example, as can be seen from Table 1 the number of distinct classes at each time period equals two. This strategy of first determining the number of latent classes for each time period before fitting the (static or dynamic) multinomial-Poisson model is adopted in the empirical application section of this paper.

Estimation and model tests

Because latent change models can be viewed as special cases of static latent-class models, the EM algorithm (Dempster, Laird, & Rubin, 1977), adapted for estimating the multinomial-Poisson latent-class model, can also be used to determine the parameters of latent change models.

Estimation. Assuming random sampling of *N* individual selection vectors over the *T* time periods, we specify the log-likelihood function as

$$\ln L = \sum_{n=1}^N \ln \Pr(\mathbf{x}_n^{(1)}, \mathbf{x}_n^{(2)}, \dots, \mathbf{x}_n^{(T)}).$$

To simplify notation, we restrict some of the following expressions to the case of three time periods and define $\mathbf{z}_n = (\mathbf{x}_n^{(1)} | \mathbf{x}_n^{(2)} | \dots | \mathbf{x}_n^{(T)})$. In the E-step of the algorithm, the posterior probabilities of person *n* belonging to classes *a*, *b*, and *c* during time periods (1), (2), and (3), respectively, is determined as

$$\pi_{abc|\mathbf{z}_n} = \frac{\pi_{abc} \Pr(\mathbf{z}_n | n \in abc)}{\Pr(\mathbf{z}_n)}.$$

Moreover, the marginal posterior probabilities for each time period are obtained from these posterior probabilities by summing over the appropriate cells,

$$\pi_{a^{(1)}|\mathbf{z}_n} = \sum_{b,c} \pi_{abc|\mathbf{z}_n},$$

$$\pi_{b^{(2)}|\mathbf{z}_n} = \sum_{a,c} \pi_{abc|\mathbf{z}_n},$$

and

$$\pi_{c^{(3)}|\mathbf{z}_n} = \sum_{a,b} \pi_{abc|\mathbf{z}_n}.$$

In the M-step, $\zeta_s^{(t)}$, $\mathbf{p}_s^{(t)}$, and π_{abc} are estimated. These parameters are estimated separately because the log-likelihood function of the "complete" data can be expressed as an additive combination of the multinomial and Poisson model part for each time period.

An estimate of π_{abc} at each M-step is obtained by

$$\hat{\pi}_{abc} = \frac{1}{N} \sum_{n=1}^N \pi_{abc|\mathbf{z}_n}. \quad (6)$$

Estimates of the choice rates at each time period are obtained by

$$\hat{\zeta}_a^{(t)} = \frac{\sum_{n=1}^N x_{n.}^{(t)} \pi_{a^{(t)}|\mathbf{z}_n}}{\sum_{n=1}^N \pi_{a^{(t)}|\mathbf{z}_n}},$$

and the conditional choice probabilities, $p_{i|a}^{(t)}$, are estimated by

$$\hat{p}_{i|a}^{(t)} = \frac{\sum_{n=1}^N x_{ni}^{(t)} \pi_{a(i)|z_n}}{\sum_{n=1}^N \pi_{a(i)|z_n} x_{ni}^{(t)}}.$$

Because these expressions can be evaluated rapidly the slow convergence behavior of the EM algorithm is more than offset by the speed of the computations at each Expectation- and Maximization-step.

Constrained estimation. The number of parameters to be estimated increases quickly with the number of time periods and latent classes. Consequently, constraints on π_{abc} , $\zeta_a^{(t)}$, and $p_{i|a}^{(t)}$ need to be considered to obtain parsimonious representations of the data. If the decision environment is stable, it is meaningful to impose equality constraints over time on the choice rates and probabilities. This is accomplished by summing over the time periods. For example, in the case of the choice rates we obtain

$$\hat{\zeta}_a = \frac{\sum_{t=1}^T \sum_{n=1}^N x_{ni}^{(t)} \pi_{a(i)|z_n}}{\sum_{t=1}^T \sum_{n=1}^N \pi_{a(i)|z_n}}.$$

Linear and loglinear constraints are imposed on π_{abc} by first computing the unconstrained estimate given by (6) and then fitting the constrained version in each M-step. This process is illustrated by the following two examples where the constrained estimates are obtained in closed form in each M-step. In these examples the unconstrained estimate is denoted by $\hat{\pi}_{abc}^*$ to distinguish it from its constrained counterpart.

To fit a nonstationary, first-order Markov model, we first compute $\hat{\pi}_{abc}^*$ by (6) and then estimate

$$\begin{aligned}\hat{\pi}_a^{(1)} &= \sum_{b,c} \hat{\pi}_{abc}^*, \\ \hat{\pi}_{b(2)|a(1)} &= \sum_c \frac{\hat{\pi}_{abc}^*}{\hat{\pi}_a^{(1)}}, \\ \hat{\pi}_{c(3)|b(2)} &= \sum_a \frac{\hat{\pi}_{abc}^*}{\hat{\pi}_b^{(2)}},\end{aligned}$$

and the constrained class-size probabilities are obtained in each M-step by

$$\hat{\pi}_{abc} = \hat{\pi}_a^{(1)} \hat{\pi}_{b(2)|a(1)} \hat{\pi}_{c(3)|b(2)}.$$

Similarly, the hypothesis of complete symmetry is fitted by

$$\hat{\pi}_{abc} = \begin{cases} \hat{\pi}_{abc}^*, & a = b = c \\ \frac{\hat{\pi}_{aac}^* + \hat{\pi}_{aca}^* + \hat{\pi}_{caa}^*}{3}, & a = b \neq c. \\ \frac{\hat{\pi}_{abc}^* + \hat{\pi}_{acb}^* + \hat{\pi}_{bac}^* + \hat{\pi}_{bca}^* + \hat{\pi}_{cab}^* + \hat{\pi}_{cba}^*}{6}, & a \neq b \neq c \end{cases}$$

Linear or loglinear constraints on π_{abc} that do not have closed-form estimates can be imposed by the iterative proportional fitting technique or Newton-Raphson methods. Mooijart and van der Heijden (1992) provide more detail about modifications of the EM algorithm when equality constraints are used in latent-class models.

Model tests. Fits are assessed by goodness-of-fit statistics and residual analyses. As overall goodness-of-fit statistic we compute Pearson's χ^2 statistic. When the expected frequencies are too small to justify the distributional assumptions of this test, the χ^2 -statistic is computed by grouping the selection patterns z_n ,

$$P^2 = \sum_{j=1}^J \frac{(f_j - \hat{f}_j)^2}{\hat{f}_j},$$

where f_j and $\hat{f}_j$ denote the observed and expected frequency of the j -th grouped selection pattern, respectively. The test statistic for grouped data does not follow asymptotically a χ^2 -distribution under the null hypothesis. However, a lower bound of the limiting distribution of P^2 is provided by a χ^2 -distribution with degrees of freedom equal to $(J - 1 - d)$, where d corresponds to the number of effective model parameters (Chernoff & Lehmann, 1954). As a result, asymptotic significance levels found from this reference distribution are too small. In addition, some care is necessary in the grouping of the selection patterns. Random groupings of the data need not converge to the grouping one gets using the true distribution function and the same grouping scheme with the consequence that standard asymptotic χ^2 -theory cannot be applied (Kulperger & Singh, 1982). In addition to testing the overall fit of the model, it is also important to examine formally or informally the fit of the model for some subsets of the data. Van Duijn and Böckenholt (1995) provide a detailed discussion of useful fit statistics and residual diagnostics for the multinomial-Poisson and related latent-class models.

The unconstrained dynamic latent class model with time-varying latent-class specific probabilities and rates and arbitrary (i.e., unconstrained) latent change requires the estimation of $(S(I + S) - 1)$ parameters. This model may prove to be too general and more parsimonious versions should be considered. In particular, the dynamic latent-Markov model with time-homogeneous choice rates and probabilities is readily interpretable and requires only $(S(I + S) - 1)$ parameters. For the comparison of nested models, large sample likelihood-ratio (LR) χ^2 -tests can be employed. An important exception constitute comparisons between models with a different number of latent classes. In this case, we do not obtain a χ^2 -statistic because the reduced model is obtained from the full model by setting parameters at their boundary values. For example, the LR χ^2 -test cannot be used to compare a static two-class model with a two-class stationary latent-Markov model because the latter model can also be viewed as a constrained static model with the number of latent classes equal to 2^T (see Table 1).

Table 2

Average prices for two brands

	Period 1	Period 2	Period 3
SW	\$0.76	\$0.60	\$0.77
CW	\$0.61	\$0.81	\$0.63

Analysis of Purchase Incidence data

In this example we analyze the purchase incidence data taken from a consumer panel of 777 households. The data set, collected over a time period of about two years, consists of choices among tuna brands belonging to the 6.5 ounce and chunk-light category. However, not every household participated in this survey for the length of this time period, and only 512 households remained in the data set after selecting the purchase data of three consecutive one-month time periods. In this application we focus on the choices between two tuna brands, Starkist in water (SW) and Chicken of the Sea in water (CW). These two brands were chosen most frequently (about 68% of the time) and, more importantly, their prices showed little variability within each time period. Table 2 displays the average prices of the two brands for each of the three time periods and shows that the brands' mean prices do not differ for the first and the third month.

To determine the number of latent classes in the data, we first analyze the data for each time period separately. Next, the total number of purchase incidences during each month is modeled and, finally, the complete data set is analyzed. Some of these analyses are redundant and they are reported only because they help illustrating various features of the static and dynamic latent-class models.

Analysis of each time period. By modeling the choice data for each time period separately, we obtain a bound on the number of latent classes necessary to account for preference heterogeneity in the data. Table 3 contains Pearson's goodness-of-fit statistics of the multinomial-Poisson model for one up to three classes. These fit statistics were computed by collapsing cells containing purchase frequencies of three and higher for the two brands. The resulting (4×4) table has 16 cells and the degrees of freedom for the various models are $16 - 3S$. According to the fit statistics in Table 3 at least two, but preferably three classes, seem necessary to account for variations in the choice probabilities and rates. Thus, a (static or dynamic) three-class model may be sufficient to model preference heterogeneity and change over the three time periods.

Analysis of choice rates. An important predictions of the multinomial-Poisson latent-class model is that brand-specific price changes affect only the choice probabilities but do not affect the choice rates. Consequently, when analyzing only the total number of purchases made within each time period we expect that the latent-class specific choice rates do not vary over time. This prediction can be tested by fitting

Table 3.

Goodness-of-fit statistics (P^2) of separate fits of the latent-class Poisson model to three time periods

LC	Period 1	Period 2	Period 3	df
1	99.7	43.3	103.8	13
2	14.8	13.1	26.6	10
3	7.7	4.6	8.3	7

latent-class Poisson models with choice rates constrained to be equal for the three time periods. The static and dynamic latent-class Poisson models are given by

$$\Pr(x_{n.}^{(1)}, x_{n.}^{(2)}, x_{n.}^{(3)}) = \sum_s \pi_s \prod_{i=1}^3 \Pr_{\Phi}(x_{n.}^{(i)}; \zeta_s),$$

and

$$\Pr(x_{n.}^{(1)}, x_{n.}^{(2)}, x_{n.}^{(3)}) = \sum_{a,b,c} \pi_{abc} \Pr_{\Phi}(x_{n.}^{(1)}; \zeta_a) \Pr_{\Phi}(x_{n.}^{(2)}; \zeta_b) \Pr_{\Phi}(x_{n.}^{(3)}; \zeta_c),$$

respectively. Because we do not expect any changes in the class-specific choice rates, it is plausible to assume that the three time periods are interchangeable with respect to the latent variations in class membership. This hypothesis was tested by imposing the constraint of complete symmetry on π_{abc} .

Table 4 summarizes the results of this analysis. The second and eighth columns contain the value of the log-likelihood functions for the one- up to four-class, latent symmetry and static Poisson models, respectively. For both models the choice rates are constrained to be equal over time. Note that the four- and three-class models have almost identical values of the log-likelihood function at the maximum. Thus, as expected from the separate analyses of the time periods, both the static and dynamic latent-class models point to a three-class solution. Pearson goodness-of-fit statistics obtained for the collapsed data table are $P^2 = 61.8$ (df = 51) and $P^2 = 84.9$ (df = 58) for the three-class latent symmetry and the three-class static Poisson model, respectively. Only the latent-class model allowing for changes in class membership provides a satisfactory fit of the data.

The LR-test statistics in Table 4 for the two hypotheses of latent symmetry and time-homogeneous choice rates were obtained by fitting the unconstrained latent-class models and comparing the solutions to their constrained counterparts. A latent-class specific rate parameter is estimated for each of the three time periods in the model with time-inhomogeneous choice rates. These parameters are constrained to be equal under the hypothesis of time-homogeneous rates. Consequently, the degrees of freedom for

Table 4

Latent-class Poisson models for total number of purchase incidences

Dynamic Poisson Models						Static Poisson Models			
LC	-logl	No. Par.	$G^2^{(a)}$	df ^(a)	$G^2^{(b)}$	df ^(b)	-logl	No. Par.	df ^(a)
1	-	-	-	-	-	-	1619.8	1	6.4
2	1542.2	5	5.9	4	7.6	4	1547.2	3	8.0
3	1530.3	12	5.2	6	7.7	17	1537.6	5	10.3
4	1530.2	23	5.2	8	9.9	44	1537.6	7	16.0

Note: ^(a) LR-statistic (G^2) with degrees of freedom (df) testing equality of choice rate parameters over time. ^(b) LR-statistic testing hypothesis of latent symmetry when choice rates are constrained to be equal over time.

the LR-test are 2.5. The degrees of freedom for testing the latent-symmetry hypothesis are $(S(S-1)(S+2)/6)$.

Table 4 shows that the hypothesis of latent symmetry is not rejected for the three- and four-class dynamic models. Moreover, in contrast to the static Poisson model, under the latent-symmetry Poisson model the hypothesis of time-homogeneous choice rates is also not rejected. These results indicate that the assumption of equal choice rates is consistent with the data only when latent change is allowed for. Moreover, because the latent change mechanism is consistent with the hypothesis of complete symmetry, class sizes do not change from one time period to the next and switches from class *a* to class *b* are as frequent as switches of the opposite direction.

Analysis of choice rates and probabilities. Table 5 contains the results of fitting the dynamic and the static multinomial-Poisson latent class model to the complete brand choice data set. In this analysis two hypotheses were tested. Because the average price of the brands changed in the second time period, there is little reason to believe that the choice probabilities stay the same over time. Consequently, we tested the hypothesis of time-homogeneous price-sensitivity. This hypothesis is equivalent to testing the constraint that the choice probabilities are equal for the first and third time period. This can be shown by expressing the log-odds of the choice probabilities as a function of price and constraining the price slope to be identical for the three time periods,

$$\ln \frac{p_{SW|a}^{(t)}}{p_{CW|a}^{(t)}} = \beta_{SC|a} + \beta_a (\$_{SW}^{(t)} - \$_{CW}^{(t)}),$$

where $\beta_{SC|a}$ represents the relative preference of SW over CW for class *a*, and the slope parameter β_a represents the class-specific price sensitivity. The second hypothesis specifies latent symmetry as the underlying mechanism for change in class membership. Support for this hypothesis was already found in the analysis of the choice rates.

The second and eighth columns of Table 5 contain the log-likelihood function

Table 5

Latent-class multinomial-Poisson models

Dynamic Multinomial-Poisson Models						Static Multinomial-Poisson Models			
LC	-logl	No. Par.	$G^2^{(a)}$	df ^(a)	$G^2^{(b)}$	df ^(b)	-logl	No. Par.	df ^(a)
1	-	-	-	-	-	-	1994.8	3	0.6
2	1903.0	9	3.2	2	8.1	4	1916.5	7	0.8
3	1886.6	18	3.8	3	10.5	17	1904.4	11	2.2
4	1882.4	31	8.5	4	10.6	44	1896.3	15	9.1

Note: ^(a) LR-statistic (G^2) with degrees of freedom (df) testing equality of choice probabilities for first and third time period. ^(b) LR-statistic testing hypothesis of latent symmetry when choice rates are constrained to be equal over time and choice probabilities are equal for the first and third time period.

values for the dynamic latent-class model (with symmetric change, and time-homogeneous price slopes and choice rates) and the static latent class models (with time-homogeneous price slopes and choice rates), respectively. Pearson goodness-of-fit tests of the collapsed data table indicate that the dynamic three-class model provides a satisfactory representation of the data ($P^2 = 57.2$, $df = 45$). In contrast, the static three-class model does not yield an adequate fit statistic ($P^2 = 83.1$, $df = 52$). LR-tests of the symmetry constraint and the hypothesis of time-homogeneous price slopes (in the fourth and sixth columns of Table 5, respectively) show that both restrictions are in agreement with the data.

The parameter estimates of the three-class symmetry model in Table 6 facilitate a straightforward interpretation of the data. Choice probabilities are equal for the first and third time period and the choice rates do not vary over time. Because of the symmetry restriction, class sizes are time-homogeneous and the transition matrix is equal for all time-period pairs. The size of the first class is small but membership is relatively stable. Members of this class buy more than members of the other classes and they are price-sensitive (i.e., they tend to buy the cheaper brand during each time period). Members of the second class are more price-sensitive than members of the first class, while members of the third class seem to be brand-loyal. However, note that the choice probabilities of the latter class are not well-determined because its members only rarely make a purchase. Overall, preferences are influenced strongly by the brands' prices. We conclude that the three classes are more a result of individual differences in purchase incidences than in preference heterogeneity.

There is no time order effect and the exchange of class members is symmetric over time regardless of whether only the total number of purchase incidences or the brand choices are taken into account. According to the estimated joint class-size probabilities, $\hat{\pi}_{abc}$, 53% of the consumers change their class membership at least once which explains the poor fit of the static three-class model. The great majority (50%) switches only between the second and third class (i.e., from medium to light buyers) and the remaining 3% switch only between the first and third class (i.e., from heavy to light

Table 6.

Parameter estimates of three-class symmetry model

LC	$\hat{p}_{SW}^{(1,3)}$	$\hat{p}_{SW}^{(2)}$	$\hat{\zeta}^{(1,2,3)}$	$\hat{\pi}^{(1,2,3)}$	$\hat{\pi}_{a b}$		
					1	2	3
1	.30	.91	2.18	.06	.85	.00*	.15
2	.11	.98	.87	.40	.00*	.58	.42
3	.84	.83	.15	.54	.02	.31	.67

Note: *These estimates took on the boundary value of 0.

buyers). Thus, there are no transitions between the first and the second class and the corresponding latent class-membership probabilities take on the boundary value of 0.

Further predictions of the latent-symmetry model are given in Tables 7 and 8. Table 7 shows that the average number of purchases per buyer, the percentage of SM and CW buyers, and the percentage of brand-loyal buyers are predicted well. Table 8 contains the predicted and observed percentages of buyers and non-buyers computed separately for the two brands. For example, 22% of the 512 households buy SW during the second time period only. CW is bought most frequently during the first (18%) and the third time period (13%). Tables 7 and 8 also show that the constraint of equal marginal distributions for the first and third time period is in good agreement with the data.

In summary, although a substantial number of consumers switches classes, we found time-homogeneous choice rates, price sensitivities, and class sizes. These results show clearly that allowing for latent change can yield a parsimonious and easy to interpret representation of recurrent choice data. As a next step, it is desirable to identify external variables that help validating the hypothesized latent change mechanism. Although this was not possible for this data set we believe the validation stage to be particularly important because of the practical and theoretical repercussions of treating class membership as either static or dynamic.

Summary and Conclusion

This paper presented dynamic latent-class models for the analysis and interpretation of stability and change in recurrent choices. The choice process at the individual level was summarized by two components, choice rates and choice probabilities for the alternatives. To account for effects of individual differences, a latent-class representation was introduced. Changes in preferences over time were represented by transitions of members from one latent class to another. The most general model allowed for latent-class specific changes in the choice rates and probabilities and a saturated (i.e., unconstrained) representation for latent switches in class membership. Several special cases were presented to obtain a parsimonious description of latent-change mechanisms. The latent Markov model, the symmetry model, and the random change model were discussed in detail. However, depending on the application, other representations of latent change may be more appropriate. In particular, the class of log-linear models

Table 7.

Predictions of three-class symmetry model

		Period 1		Period 2		Period 3	
		SW	CW	SW	CW	SW	CW
% Buyers	Observed	14	30	32	4	14	25
	Predicted	14	27	34	4	14	27
Purch. per buyer	Observed	1.1	1.5	1.5	1.1	1.1	1.5
	Predicted	1.1	1.5	1.5	1.1	1.1	1.5
100% loyal	Observed	23	21				
	Predicted	25	20				

Table 8

Observed and Predicted Purchase Percentages

t_1	t_2	t_3	SW		CW	
			pred	obs	pred	obs
0	0	0	50	53	55	55
0	0	1+	7	6	16	13
0	1+	0	24	22	2	1
0	1+	1+	4	4	1	1
1+	0	0	7	7	16	18
1+	0	1+	1	2	9	11
1+	1+	0	4	4	1	1
1+	1+	1+	1	2	1	1

Note: The first three columns refer to the purchase patterns for the three time periods. The grouped selection patterns are denoted by 0 (no purchase) and 1+ (at least one purchase).

facilitates testing a rich set of hypotheses about the underlying change mechanisms. Nonstationarity resulting from changes in the decision-making environment can also be accounted for by re-parametrizing choice probabilities and rates as a function of external variables.

An important issue that should be addressed in future research is the treatment of missing data, for example, as a result of attrition. Because missing data are unavoidable in longitudinal studies, the approach presented in this paper to consider only panel members with complete responses may not be satisfactory in some applications. Specifically, if the mechanisms that lead to the missing data are an integral part of the choice behavior under study, they need to be explicitly taken into account in an analysis of the data.

In conclusion, we note that the use of latent change mechanisms is not restricted to the multinomial or the multinomial-Poisson latent-class models. For example, only minor modifications are necessary to allow for class-membership shifts in the various mixture models for recurrent choice data discussed in Böckenholt (1993). The advantages of such extensions were illustrated by the presented application: Dynamic latent-class models provide considerable flexibility and parsimony in modeling the separate effects of taste heterogeneity and preference changes in longitudinal choice data.

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Omar Lizardo (olizardo)
Sociology
810 Flanner
Notre Dame, IN 46656